MLP - Assignment 1

23/10/26

Q.1) An online store offers discounts based on the purchase amount:
10% Discount for purchase between $100 and $500
20% Discount for purchases above $500
no Discount for purchases below $100

Execution:

```
amount = int(input("enter amount:"))
if amount < 100:
        Discount = 0
elif amount >= 100 and amount <= 500:
        discount = amount * 10/100
else:
        discount = amount * 20/100
final = amount - discount
print("discount:", discount)
print("final amount:", final)
```

Output:

```
enter amount : 700
discount     : 140.0
final amount : 560.0
```

2) create a program that simulates a traffic light. the user inputs one of the colors: Red, yellow, or Green.
Red, display "stop"
yellow, display "Ready to go"
Green, "Go", for invalid input, display "invalid"

execution

```
color = input("Enter traffic light color:")
if color == "Red":
    print("stop")
elif color == "yellow":
    print("Ready to go")
elif color == "Green":
    print("Go")
else:
    print("invalid color")
```

output:

```
enter traffic light color: green
go
```

3) A school uses the following grading system:
Marks >= 90 : Grade A
Marks >= 75 and < 90 : Grade B
Marks >= 50 and < 75 : Grade C
Marks < 50 : fail

Execution:

```
marks = int(input("Enter marks:"))
if marks >= 90:
    print("Grade A")
elif marks >= 75 and marks < 90:
    print("Grade B")
elif marks >= 50 and marks < 75:
    print("Grade C")
elif marks >= 0 and marks < 50:
    print("fail")
else:
    print("not allowed student")
```

output:

```
enter marks = 56
Grade C
```

4: write a program that takes an integer as input.

whether the number is odd or even
whether the number is divisible by 5

Execution:

```
num = int(input("enter a number:"))
if num % 2 == 0:
    print("even number")
else:
    print("odd number")

if num % 5 == 0:
    print("divisible by 5")
else:
    print("not divisible by 5")
```

output:

```
enter a number : 60
  even number
divisible by 5
```